

Anurag Bhatt

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Data Scientist with **7+ years of experience**, specializing in **Machine Learning, Deep Learning, NLP and Generative AI**. Proven ability to leverage these skills across diverse industries, including **US healthcare, insurance, manufacturing, finance, and Automobile**. Proficient in cloud platforms like **GCP, AWS ML tools, and Azure ML**. Dedicated professional committed to team success through **hard work, attention to detail**, and strong organizational skills. Seeking a role in a forward-thinking company that values best practices and offers diverse customer projects.

WORK EXPERIENCE

Senior Consultant, Capgemini

May 2025 – Present | Gurgaon, Haryana

- **Improved search accuracy by 40%** by building a retrieval system that prioritizes context, specifically using summaries and FAQs to give users more relevant answers.
- **Cut API operational costs by 60%** by creating a smart routing system that automatically picks the most cost-effective model for each specific task.
- **Solved the issue of "vague" user questions** by adding a hidden refinement step that cleans and expands queries to improve search intent before processing.
- **Automated the processing of 10,000+ data chunks** for a user base of over 1,000 people, streamlining the entire document upload and ingestion workflow.
- **Fixed text extraction failures in complex PDFs** by using AI vision to accurately pull data from difficult layouts and tables that standard tools missed.
- **Built a code-risk analysis tool for GitHub** that uses Tree-sitter and a Gremlin graph database (Azure Cosmos DB) to map out how code changes impact the rest of the system.
- **Simplified the PR review process** by designing a prompt-chaining strategy that turns messy file diffs into clear, high-level architectural impact reports.

ML Engineer, OUTBOUND AI TECHNOLOGIES PRIVATE LIMITED

December 2023 – May 2025

- Spearheaded **comprehensive claims analytics** for 100+ payers, utilizing **statistical analysis** and **AI-driven root cause identification** to optimize processing efficiency using python.
- Engineered a **scalable data extraction framework** for diverse claim types, implementing **configurable templates** to accommodate various document formats and improve accuracy.
- Leveraged cutting-edge **Generative AI models**, including **GPT, llama, etc** to extract insights from call transcripts and generated **actionable summaries** for enhanced customer intelligence using python libraries.
- Developed and applied **machine learning algorithms** to differentiate between payer-specific and systemic issues, enabling **targeted problem-solving strategies**.
- Created impactful **data visualizations** and **predictive models**, translating complex findings into **strategic recommendations** for senior management decision-making.
- Improved **operational efficiency** and **payer experience** through proactive identification and resolution of **AI-driven claim processing discrepancies**.

Data Scientist, TATA CONSULTANCY SERVICES LIMITED

January 2019 – December 2023 | Pune, India

- Collaborated with the **Research and Innovation Team** on multiple **analytics projects**
- Contributed to a team-oriented environment, focusing on the delivery of **high-quality programs and applications** for the **Artificial Intelligence and Analytics industry**
- Worked on **cutting-edge patents** and **R&I modules**, promoting **industry-first solutions**
- Developed **end-to-end applications** for:
 - **Customer Experience Analysis**
 - **Skill-Based Routing**
 - **Natural Language Processing** and Forecasting Solutions
 - **Generative AI Solutions** for Finance

- Maintained consistent communication with team members and other stakeholders, ensuring smooth coordination and efficient delivery of results

Research Assistant, TATA CONSULTANCY SERVICES LIMITED

July 2018 – December 2018 | Chennai, India

- It's been great learning experience, have also worked on vast set of modules in field of Research and Innovation in TCS IGNITE Lab as Research Assistant.
- Developed a comprehensive Website Vulnerability Scanner
- Capable of detecting SQL Injection, Cross-site Scripting, etc.
- Leveraged Python programming language.
- Utilized Flask framework for efficient backend development.

PROJECTS

Claim Status Reporting

- Led **comprehensive claims analytics** for a diverse portfolio of **100+ payers**, leveraging **statistical analysis** to identify **performance trends** and **AI-generated root causes** of issues, while adhering to strict **HIPAA compliance** for **restricted PHI data** in US healthcare.
- Developed and implemented **sophisticated data extraction algorithms**, optimizing **configuration processes** and enhancing **speech-to-text accuracy** for improved claims processing, ensuring **secure handling of sensitive patient information**.
- Utilized **advanced statistical methods** to differentiate between **payer-specific challenges** and **systemic issues**, ensuring **targeted problem-solving** approaches within a **highly regulated data environment**.
- Designed and executed **data verification protocols** to maintain high **customer data accuracy** and **PHI security**, critical for reliable analytics and decision-making in healthcare.
- Proactively identified and resolved **AI-driven claim processing discrepancies**, significantly improving **operational efficiency** and enhancing overall **payer experience**, all while safeguarding **sensitive health information**.
- Engineered a robust **data extraction framework** for various claim types, including **Explanation of Benefits (EOB)**, ensuring accurate data allocation and document assignment within the product portal, with strict **access controls** for PHI.
- Developed **configurable templates** to streamline data extraction processes, accommodating diverse **payer-specific EOB document formats** and improving scalability, while maintaining **HIPAA compliance**.
- Leveraged **state-of-the-art Generative AI models**, including **GPT** and various **open-source alternatives**, to extract valuable insights from **call transcripts** and generate **comprehensive summaries** for enhanced customer intelligence, implementing **rigorous data anonymization** techniques to protect patient privacy.

Financial Q&A System: Accurate Customer Document Analysis

- Successfully developed and implemented a robust **Question and Answering** system with **generative AI** to extract accurate financial information from customer documents.
- Proficiently utilized tools such as **llama_index**, **langchain**, and the open-source **LLM** model **Flan-T5-large** in conjunction with **Azure ML** and **Azure Open AI**.
- Achieved precise and detailed explanations, ensuring **high-quality results**.

Cobuddy

- Designed a **"Context-Aware" search** that looks at summaries and FAQs, making answers **40% more accurate**.
- Built a **smart routing system** that picks the cheapest model for the job, **slashing API costs by 60%**.
- **Fixed "vague" user questions** by adding a hidden step that cleans up and **expands queries** before searching.
- **Automated the document upload process** to handle 10,000+ chunks of data for **over 1,000 active users**.
- Solved complex **PDF reading issues** by using AI vision to extract text from **difficult layouts and tables**.

Change Analyzer

- **Built a GitHub-integrated analysis tool** using the **OpenAI API** to summarize code changes and flag risks during the PR process.
- **Developed a multi-language parser** with **Tree-sitter** to traverse **Abstract Syntax Trees (AST)** and map complex code dependencies.
- **Designed a Gremlin-based graph schema** in **Azure Cosmos DB** to model relationships between functions, classes, and source files.
- **Engineered a prompt-chaining strategy** that synthesizes individual file diffs into **coherent impact analysis reports** for large PRs.

- **Mapped function signatures and call graphs** to create a queryable knowledge graph, improving the **accuracy of automated code reviews**.

SKILLS

Natural Language Processing (NLP) • **Flask** • **Python** • **Machine Learning** • **AWS** — Sagemaker, EC2 Instance, SSM, Lambda Function, SNS, S3 Bucket, Comprehend • **Scikitlearn** • **Numpy** • **TensorFlow** • **Transformers** • **Deep Learning** • **Docker** • **Git** • **Generative AI** • **Transformers** • **Llama Index** • **Langchain** • **Microsoft Azure** — Cognitive Search, ML Studio, Azure Open AI • **Vector Database** — Qdrant • **FastAPI** • **Transformers** • **Embeddings** • **LLM's** — GPT, LLama, Gemma • **Jira** • **Git** • **Open AI API** • **Azure** — Cosmos DB, Gremlin API, Azure OpenAI Service, Azure Web Apps, Azure ML, MSAL • **Github Actions** • **Knowledge Graphs** • **Prompt Engineering** • **Langchain** • **Microsoft 365 API** • **MongoDB** • **Neo4j** — Graph Data Science, Vector Search • **Docker** — Docker Compose • **Rasa NLU**

EDUCATION

Master in Computer Applications, <i>SASTRA Deemed University</i>	2018 – 2021
Bachelor of Science (Information Technology), <i>Hemvati Nandan Bahuguna Garhwal University</i>	2015 – 2018